

Elecard YUV Viewer 2024

User Guide

User Guide Notices

Elecard YUV Viewer 2024 (4.4) User Guide

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1. Introduction

1.1 Preface

Elecard YUV Viewer is a powerful tool designed for professionals and prosumers in the video compression field to view YUV data sequence in YUV video files, compare files against one another to find out if binary images match, and view the result of comparison. The application also allows the user to calculate quality metrics, such as PSNR, APSNR, MSE, MSAD, Delta, SSIM, NQI, VIF, VQM, VMAF and VMAF phone. It has been designed and implemented for reverse engineering and analysis.

1.2 Describing Elecard YUV Viewer

The following section defines the specifications and features of Elecard YUV Viewer. It includes definitions and descriptions of the supported file types.

1.2.1 Supported File Types

- I444
- I422
- IYUV
- NV12
- NV21
- P420
- P444
- RGB24
- RGB32
- UYVY (Y422, UYNV, HDYC)
- V210
- V400
- V444
- Y42B
- YUY2 (YUNV, V422, YUYV)
- YV12
- YV16
- YVYU

Note: If the opened file is not YUV (for instance RGB), it is converted into the YUV color space.

For detailed information on video formats please refer to <http://fourcc.org>.

1.2.2 Metric Types

- **PSNR** – Peak Signal-to-Noise Ratio.
- **APSNR** – average values of PSNR results calculated for a sequence and based on mean values of MSE results.
- **VQM** – Video Quality Measurement Techniques – a metric designed to calculate video quality and measure perceptual effect of video distortions.

- **SSIM** – Structural SIMilarity. This metric is based on measuring three components (luminance similarity, contrast similarity and structural similarity) and combining them into result value.
- **DELTA** – the value of this metric is the mean difference of the color components in the correspondent points of image.
- **MSE** – Mean-squared-error based metric.
- **MSAD** – the value of this metric is the mean absolute difference of the color components in the correspondent points of image.
- **NQI** – New Quality Index - is a metric designed to calculate video quality by combining three components: correlation loss, luminance and contrast distortion.
- **VIF** – Visual Information Fidelity is a full reference image quality assessment index based on natural scene statistics and the notion of image information extracted by the human visual system.
- **VMAF** – Video Multimethod Assessment Fusion is an objective full-reference [video quality](#) metric developed by [Netflix](#) in cooperation with the [University of Southern California](#).
- **VMAF phone** – a metric designed to predict the quality of videos displayed on mobile phones.

1.2.3 Features

- Cross-platform: Windows, macOS, Linux;
- Off-line activation includes PC-based and dongle-based activation types;
- Displays YUV sequences;
- Compares files for binary image matching;
- Displays the comparison results;
- Calculates quality metrics (PSNR, APSNR, MSE, MSAD, Delta, SSIM, NQI, VIF, VQM, VMAF, and VMAF phone);
- New file types supported: I444, I422, IYUV, NV12, NV21, P420, P444, RGB24, RGB32, UYVY (Y422, UYNV, HDYC), V210, V400, V444, Y42B, YUY2 (YUNV, V422, YUYV), YV12, YV16, YVYU;
- Sharing comments between application instances and/or applications of Elecard StreamEye Studio set;
- Synchronization between applications of Elecard StreamEye Studio set (Binding mode);
- Automated analysis of video arrays via the command line tool.

1.3 Using This Guide

1.3.1 Purpose

This guide is intended to help the user utilize Elecard YUV Viewer. It describes the YUV Viewer GUI, settings and functions and provides instructions for using the viewer.

1.3.2 Topics Covered

The following lists the topics covered in this document:

- **Section 1: Introduction** – provides a general overview of Elecard YUV Viewer and describes the purpose of the document and its contents.
- **Section 2: Getting Started** – describes how to install, uninstall and run Elecard YUV Viewer.
- **Section 3: Using YUV Viewer** – describes the Elecard YUV Viewer GUI and provides instructions for comparing video streams and viewing the results of comparison.

- **Section 4: Command Line Tool** – describes the ElecCard YUV Viewer Command Line tool and provides the commands example.

1.4 System Requirements

1.4.1 Hardware Requirements

- 8 GB RAM for HD video (or higher).
- 16 GB RAM for UHD video (or higher).

1.4.2 Software Requirements

- macOS 10.13 High Sierra — Mac OS 14 Sonoma

1.5 Licensing and Technical Support

By installing, copying, or otherwise using the SOFTWARE PRODUCT or any UPDATES, you agree to be bound by the terms of the "ElecCard" End-User License Agreement ("EULA"). This EULA is a legal agreement between you (either an individual or a single entity) and ElecCard for the "ElecCard" software product(s) accompanying this EULA, which include(s) computer software and may include "online" or electronic documentation, associated media, and printed materials ("SOFTWARE PRODUCT").

For technical support, please contact the ElecCard Technical Support Team: tsup@eleccard.com.

For sales and licensing information contact the ElecCard Sales Department: sales@eleccard.com.

1.6 Activating ElecCard YUV Viewer

1.6.1 Sentinel License Protection

Make sure that Sentinel License Manager software has been installed on your computer. License Manager is provided within the product installation pack for Windows OS and macOS and installed automatically.

License Manager is provided as a separate installation pack for Linux OS and should be installed manually.

Please note, macOS reinstallation will make the activated product license invalid.

To check that License Manager is properly installed, go to Sentinel Admin Control Center at <http://localhost:1947>. If you see your license information, the installation is successfully completed.



Figure 1. The License Manager with license information

License Manager allows configuring, controlling, monitoring and looking through a list of available licenses. For more details on License Manager operation click the Help tab.

Note, each page contains the Help tab related to this page only.

You can activate Elecard YUV Viewer offline via following steps:

For Elecard YUV Viewer activation, a C2V (client-to-vendor) file should be created. To create the C2V file follow the steps:

1. Open the **Application** (YUVViewer) tab and select **Preferences** in the drop-down menu.
2. A dialog window should open. Open the **License** tab.
3. Select HASP-SL for PC-based or HASP-HL for dongle-based activation in the *Offline activation with license files* field. Note, if you select a PC-based (HASP-SL) activation, full featured activated Elecard YUV Viewer will run only on this particular hardware. If you have Elecard dongle, plug it into a free USB port on your computer and select dongle-based (HASP-HL) activation. It will be possible to use Elecard product on any hardware with the dongle plugged in.

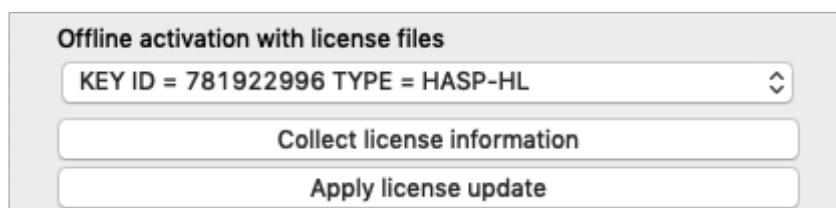


Figure 2. Offline activation with license files

4. Click the **Collect license information** button. The Save File dialog box should open. Click **Save** and save the C2V file to a required directory. Send the saved C2V file to Elecard Sales at sales@elecard.com.

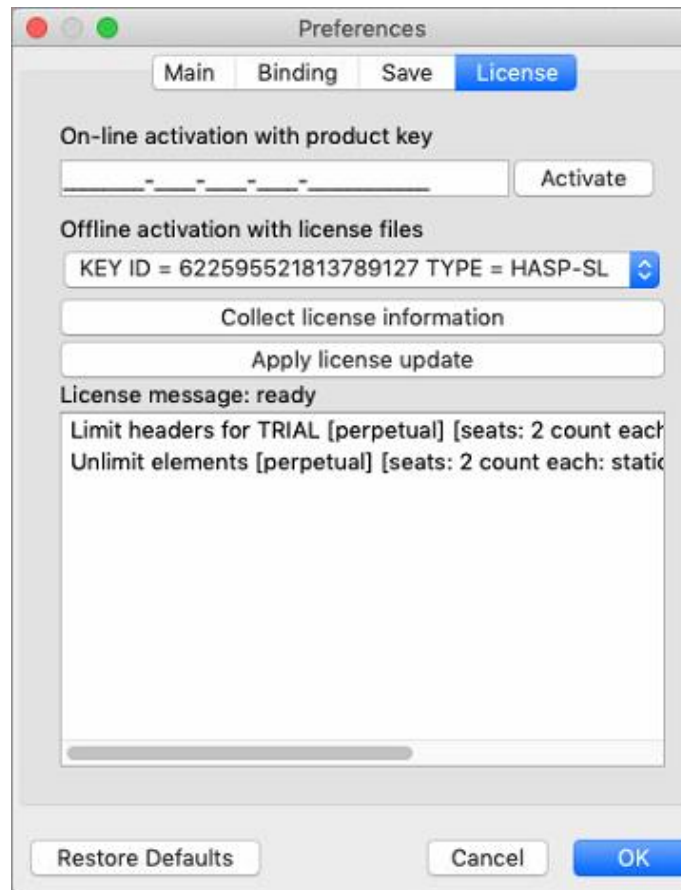


Figure 3. Elecard YUV Viewer GUI – Creating a C2V file: License tab

5. After receiving the V2C file containing license information back from Elecard Sales save the received file to a required directory. Open the **Application** (YUVViewer) tab and select **Preferences** in the drop-down menu again. Open the **License** tab and click **Apply license update** (see Figure 4). The file selection dialog box should appear. Choose the received V2C file and click **OK**. Your application is successfully activated.

2. Getting Started

The following section details the procedures for installing, registration, uninstalling, and running Elecard YUV Viewer.

2.1 Installing Elecard YUV Viewer

Elecard YUV Viewer is supplied as part of Elecard StreamEye Studio or as a standalone application. After installing Elecard StreamEye Studio or Elecard YUV Viewer on macOS, all applications are located in the Applications folder:

- Applications → Elecard StreamEye Studio 2024;
- Applications → Elecard YUV Viewer 2024.

The console is located in: Users → bin → YUVconsole → yuvviewer-cli.

The documents are located in: Library → Application Support → Elecard → YUV Viewer 4 → Elecard YUV Viewer 2024.pdf.

2.2 Uninstalling Elecard YUV Viewer

Elecard YUV Viewer can be uninstalled as part of Elecard StreamEye Studio or as a standalone application. To uninstall the application in macOS, move the required application to the trash icon.

Elecard StreamEye Studio or Elecard YUV Viewer applications are located in the Applications folder:

- Applications → Elecard StreamEye Studio 2024;
- Applications → Elecard YUV Viewer 2024.

The console is located in: Users → bin → YUVconsole.

The documents are located in: Library → Application Support → Elecard.

2.3 Running Elecard YUV Viewer

To run Elecard YUV Viewer, click *Applications → Elecard StreamEye Studio 2024 → Elecard YUV Viewer 2024*

To run Elecard YUV Viewer, click *Applications → Elecard YUV Viewer 2024*

To run Elecard YUV Viewer double click on the application icon, and the application will automatically run.

3. Using Elecard YUV Viewer

3.1 Introduction

The following section describes the Elecard YUV Viewer GUI (graphical user interface), its features, and instructions for viewing YUV sequences, comparison of two files (file **A** and file **B**), and viewing the results of comparison.

3.2 Describing Elecard YUV Viewer GUI

The following section describes the Elecard YUV Viewer GUI.

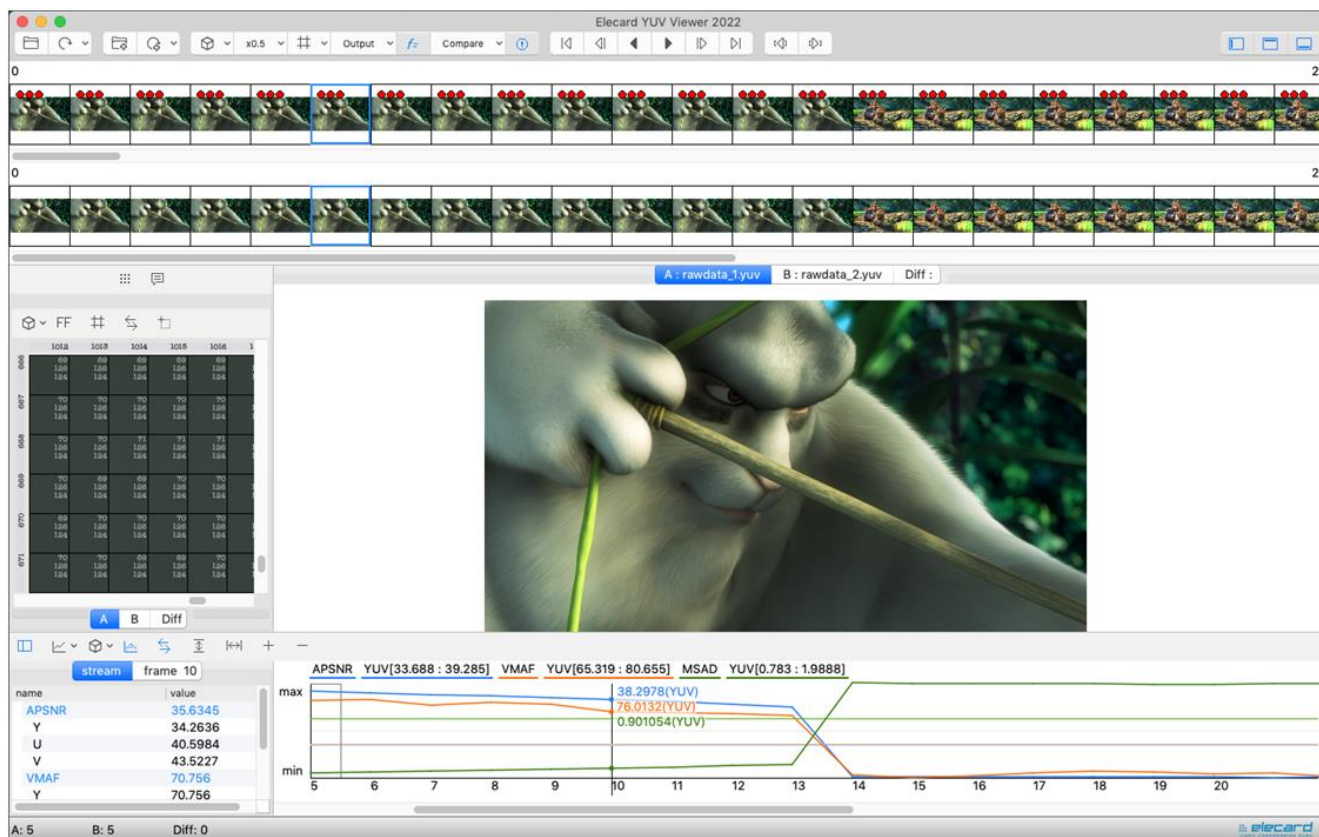


Figure 4. Elecard YUV Viewer GUI

The following table describes the **Main** menu commands and the corresponding toolbar buttons.

Table 1. Elecard YUV Viewer Menu and Toolbar

YUVViewer		
About		Shows the About window.
Preferences		Opens the program settings dialog (Options dialog).
Quit		Exits the application.
File		
Open A		Opens file A.

Reopen A		Opens the recently opened file A.
Reopen Last A		Reopens the last opened file A.
Close A		Closes file A.
Open B		Opens File B.
Reopen B		Opens the recently opened file B.
Reopen Last B		Reopens the last opened file B.
Close B		Closes file B.
Save		Saves metrics data, YUV data and image (A, B, Difference).
Binding		Enables or disables the Binding mode: Leader – allows sending comments, commands on interface or positioning changes to other applications; Follower – allows receiving comments, commands from other applications.
Open Comments...		Opens comments from the XML file.
Save Comments...		Saves comments to the XML file.
View		
Color Space		Selects color space to be displayed: YUV, Y, U, V.
Scale	x1	Sets the video scaling coefficient: x0.125, x0.25, x0.5, x1, x2, x4, x8.
Grid		Displays grid: none, 4x4, 8x8, 16x16, 32x32.
Show Error		Shows errors on YUV components (red circle corresponds to unmatched data between A and B streams, green circle corresponds to the matched data).
Show All		Shows all panels.
Hide All		Hide all panels.
Panels		Allows displaying or activating the Navigation, Graphics, Zoom panels.
Output		Output – shows and activates the current window (A / B / Difference).
Difference		Enables/disables the difference mode.

Difference Type		Enables/disables the comparison of A stream with B stream. The combo-box Difference Mode is displayed. The following Difference modes are available: • Compare – pixel-by-pixel comparison of two frames. If two pixels are matched, the result is 0. Otherwise the result is equal to the maximum value. • Subtraction – pixel-by-pixel subtraction. The result is equal to the difference value. • Temperature – absolute difference (pixel-by-pixel subtraction), results are highlighted (black-blue-green-red), where black corresponds to zero difference, blue corresponds to slight difference, green corresponds to medium difference and red corresponds to the biggest difference. • PSNR – the result is the PSNR values (from 0 to 99.99) for corresponding CU partitioning blocks, where black color corresponds to 0 value, and white color corresponds to the value of 99.99 and higher. • PSNR Clip – the same as the PSNR, but minimum result value is the minimal PSNR among all blocks in the frame and maximum – the maximal PSNR, respectively. Black color corresponds to minimal value, white color corresponds to maximal value, and blended colors correspond to intermediate values.
Navigation		
Go to Begin	●	Go to beginning - sets the current position to the first frame in the stream.
Step Backward	◀	Previous - sets the current position to the previous frame.
Play Backward	◀	Play Backward - starts the stream reverse playback.
Play forward	▶	Play - starts and stops the stream playback.
Go to End	▶	Go to end - sets the current position to the last frame in the stream.
Step Forward	▶	Next - sets the current position to the next frame.
Offset Decrement / Increment	◀ / ▶	Decreases / increases offset of the current frame in the file B relatively to the current frame in the file A.
Help		
Search		Opens the Search dialog with search options.
Feedback	💬	Sends a feedback on YUV Viewer via e-mail with the following subjects: <i>Submit an Idea, Ask a Question, Report a Bug.</i>
Contents...		Opens the current User Guide in .pdf format
Presentation mode		
Left Panel	☐	Hides/shows the Zoom panel at the left of the application window.
Bottom Panels	☐	Hides/shows the Navigation panel at the top of the application window.
Right Panel	☐	Hides/shows the Graphics panel at the bottom of the application window.

3.2.1 Options Dialog

To configure file settings, binding and saving modes, open the **Options** dialog by selecting the **Preferences** menu command from the **YUVViewer** tab.

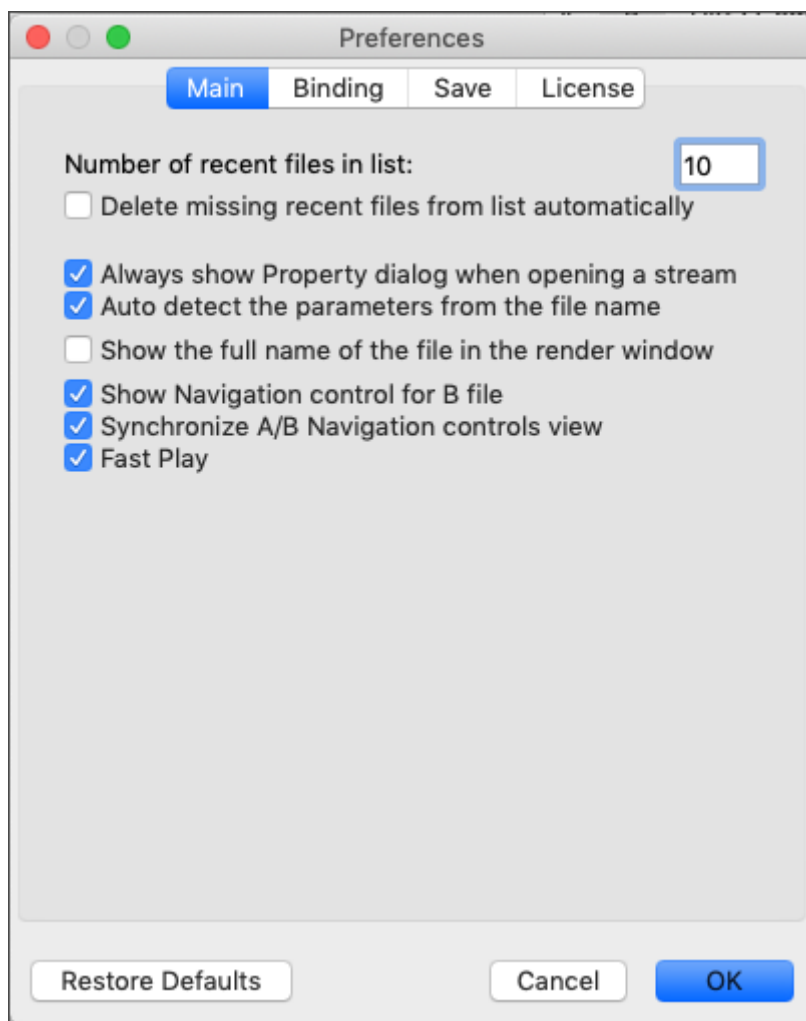


Figure 5. Elecard YUV Viewer GUI – Options Dialog – Main Tab

The following table describes the Main tab in the Options dialog.

Table 2. Elecard YUV Viewer – Options dialog – Main tab

Control	Function
Number of recent files in list	Shows a number of recent files being saved in a list.
Delete missing recent files from list automatically	Allows deleting missing files from the recent list automatically when opening a file from the list.
Always show Property dialog when opening a stream	Enables/disables the Property dialog displaying when opening a stream. If the option is disabled, the stream is opened with the last used width, height, and type parameters.
Auto detect the parameters from the file name	Enables/disables auto detection of the stream parameters using the file name. If the file name contains a substring WxH (where W and H – certain numbers), the specified numbers will be used automatically as the with and height values.
Show the full name of the file in the render window	Shows the file name in the render window.
Show Navigation control for B file	Enables/disables the thumbnails line for the B file
Synchronize A/B Navigation controls view	Allows synchronizing view for A/B Navigation controls

Fast play	Allows playing back a media file with maximum speed (if enabled) or equalizes playback speed at regular intervals: number of frames per second (if disabled).
------------------	---

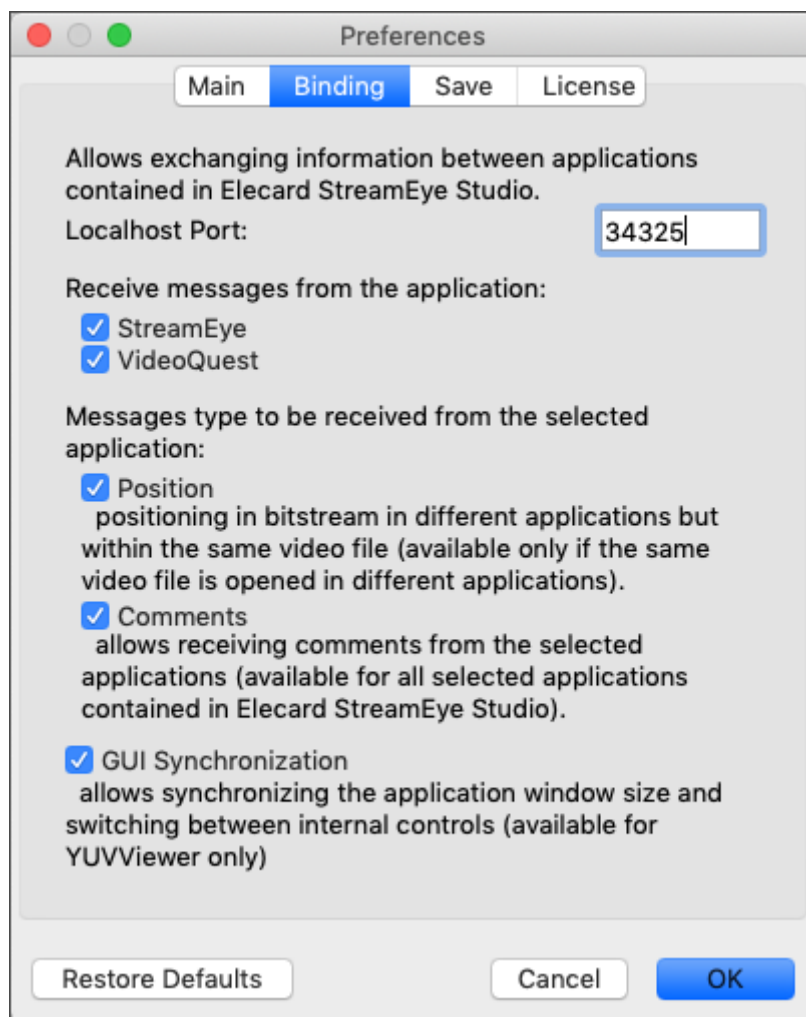


Figure 6. Elecard YUV Viewer GUI – Options Dialog – Binding Tab

The Binding tab contains settings that allow configuration of the interaction between Elecard StreamEye Studio applications. In the Binding Mode, exchanging information between applications is provided via a specified port. By activating checkboxes, you can select which applications should send messages and set messages type that should be communicated. Additionally, the synchronization of applications window size and switching between controls can be enabled.

Note: The Restore Defaults button allows you to reset set options to their original values. Functionality of this button is not applicable to the License tab.

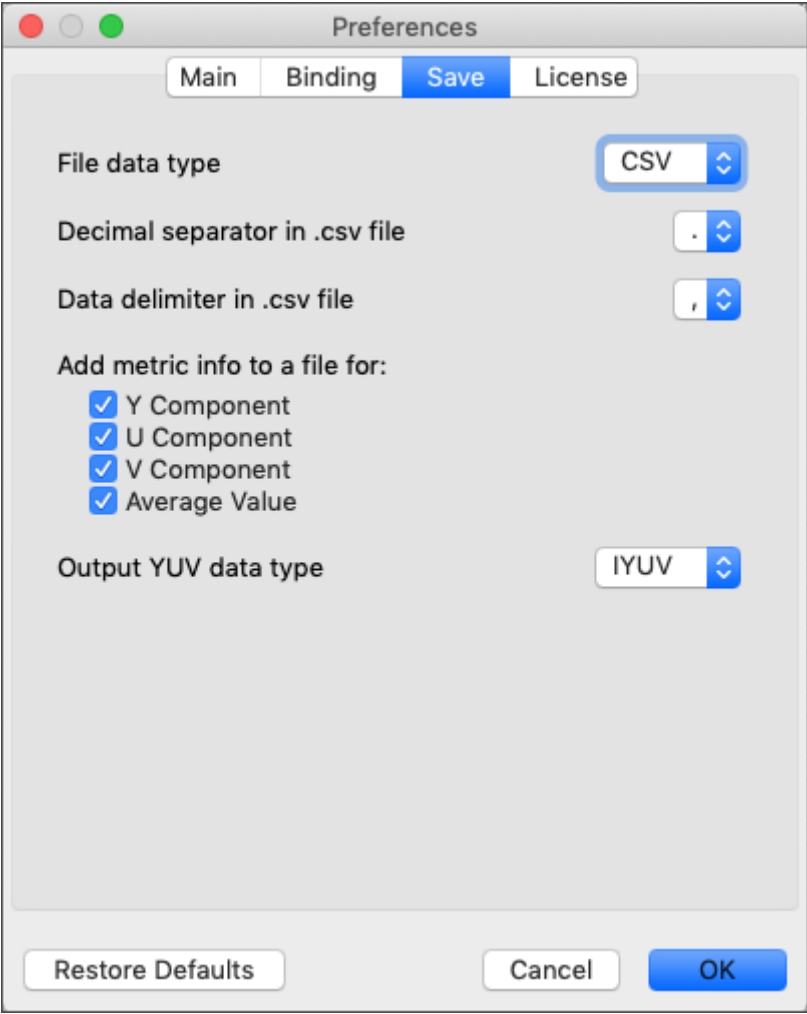


Figure 7. Elecard YUV Viewer GUI – Options Dialog – Save Tab

To configure the saving options, open the **Save** tab from the Option dialog and see the table below.

Table 3. Elecard YUV Viewer – Options dialog – Save tab

Control	Function
File data type	Allows selecting file data type: CSV, JSON.
Decimal separator in .csv file	Allows selecting a decimal separator.
Data delimiter in .csv file	Allows selecting data delimiter.
Add metric info to a file for	Allows adding metric info for the selected parameters when saving: Y component, U component, V component or average value.
Output YUV data type	Allows saving YUV data in one of the following formats: I444, I422, IYUV, NV12, NV21, P420, P444, RGB24, RGB32, UYVY (Y422, UYNV, HDYC), V210, V400, V444, Y42B, YUY2 (YUNV, V422, YUYV), YV12, YV16, YVYU.

3.2.2 Defining YUV Viewer Parameters

The section gives an overview of the **Property** dialog window displayed when the user opens a file to be analyzed.

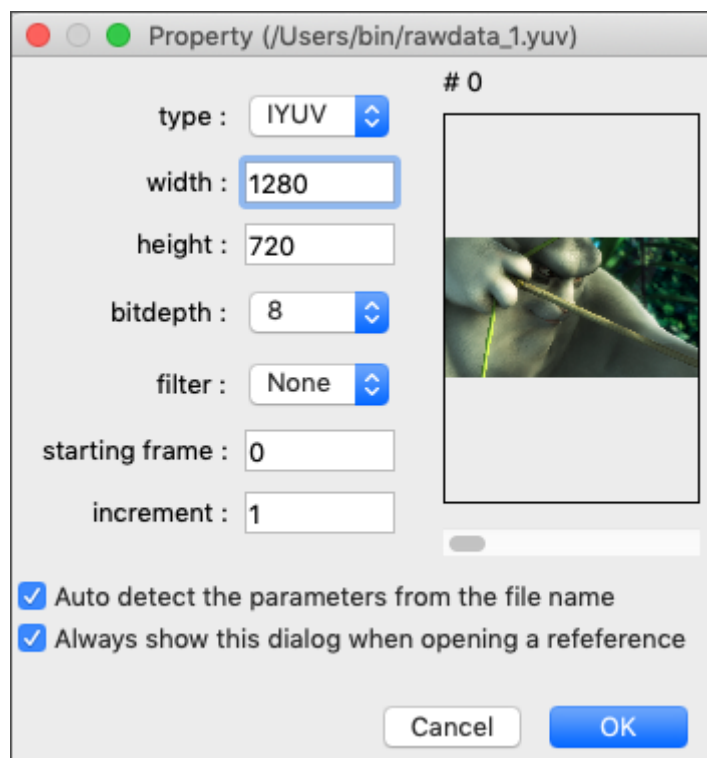


Figure 8. Elecard YUV Viewer – Property Dialog

The following table describes the **Property** dialog controls.

Table 4. Elecard YUV Viewer Menu and Toolbar

Control	Function
Type	Specifies the type of the YUV data to be analyzed.
Width	Specifies the width of the frame for the file to be analyzed.
Height	Specifies the height of the frame for the file to be analyzed.
Bitdepth	Allows a user to specify values for some metric Types. For several Types, the value is not editable.
Filter	Selects the preprocessing filter.
Starting frame	Allows specifying a starting frame (from the beginning of a reference stream).
Increment	Specifies an increment value, i.e. coefficient used for calculation of a successive frame number intended for further analysis. 1 is set by default.
Auto detect the parameters from the file name	Enables/disables auto detection of the stream parameters using the file name. If the file name contains a substring WxH (where W and H – certain numbers), the numbers will be used automatically as the with and height values.
Always show when this dialog when opening a reference	Enables/disables the Property dialog displaying after the stream opening. If the option is disabled, the stream is opened with the last specified width, height, and type parameters.
Ok	Confirms the specified parameters and closes the dialog.
Cancel	Closes the Dialog. The file is not opened.

When a stream or streams are open, the **Output Window** is displayed: the selected frames or difference is shown in the corresponding windows.

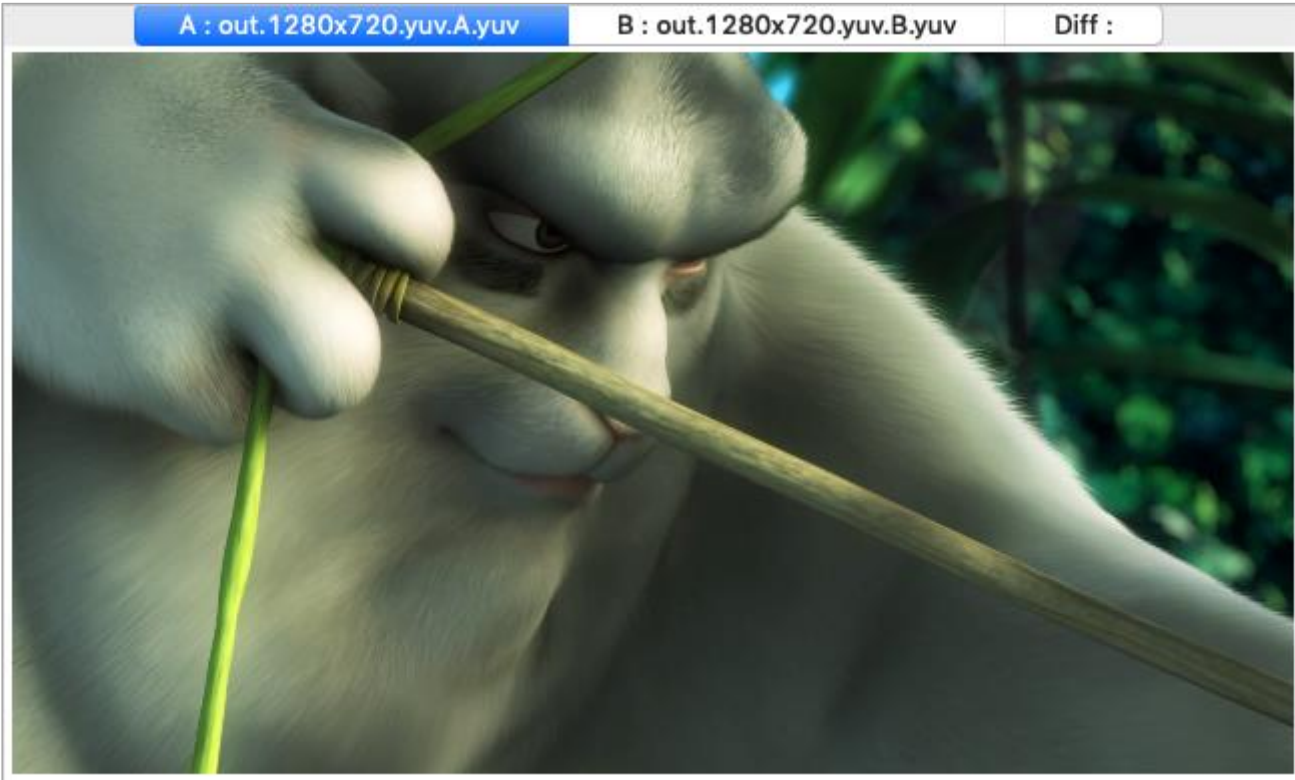


Figure 9. Elecard YUV Viewer – Output Window

3.2.3 Zoom Panel

The **Zoom** panel displays pixel YUV values for stream A, stream B and difference.

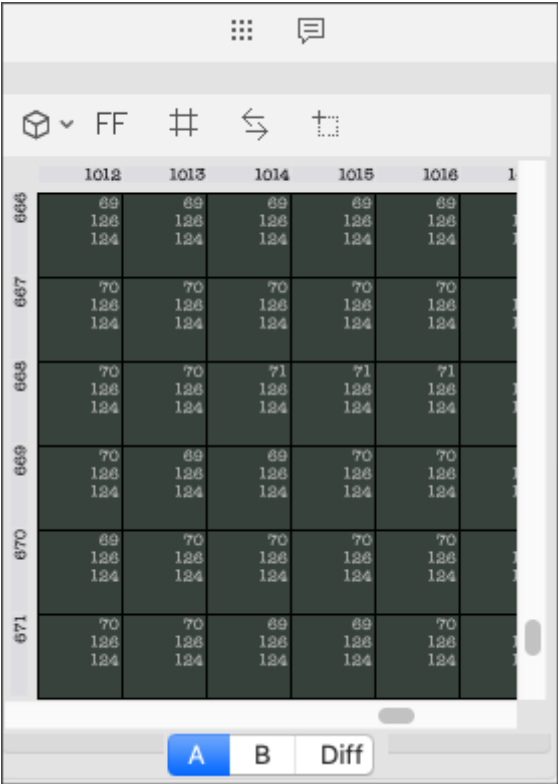


Figure 10. Elecard YUV Viewer – Zoom Panel

The following table describes the **Zoom** panel controls.

Table 5. Zoom Panel Controls

Control	Description
	Selects YUV color space to be displayed. The background color corresponds to the pixel color.
FF	Displays data in decimal\hexadecimal mode.
#	Displays grid.
	Synchronizes all other windows with the marked position.
	Allows displaying the zoomed region in the output panel.

3.2.4 Graphics Panel

The Graphics Panel is located at the bottom of the application window.

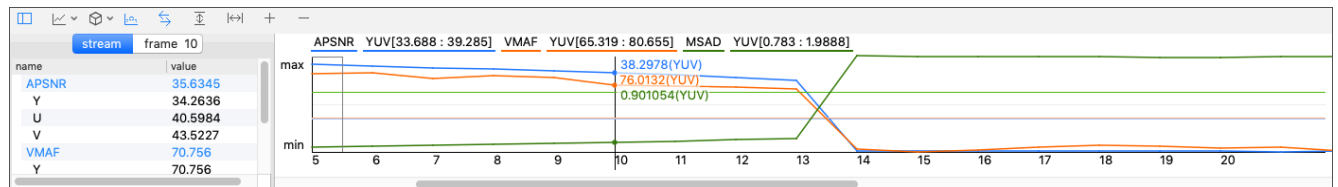


Figure 11. Elecard YUV Viewer – Metrics Window

The following table describes the **Graphics** panel controls.

Table 6. Graphics Panel Controls

Control	Description
	Shows or hides information panel displaying the selected metrics and their values.
	Shows the selected metric types which are displayed in the graphics. To enable or disable metrics calculation, set checkboxes.
	Allows selecting YUV components for visualization results.
	Allows displaying results in the Info panel when moving a mouse pointer if enabled or displaying results only when positioning a mouse cursor on a frame if disabled.
	Allows setting vertical borders on a graph
	Shows average values of the selected metrics.
	Fitting the current range.
+ -	Scaling.

Please note, that only the selected metrics are calculated. Calculation of a metric starts as soon as the metric is selected.

If a metric has not been calculated for some frames, the uncalculated segments are displayed as dashed lines, but the calculated segments are displayed as solid lines.

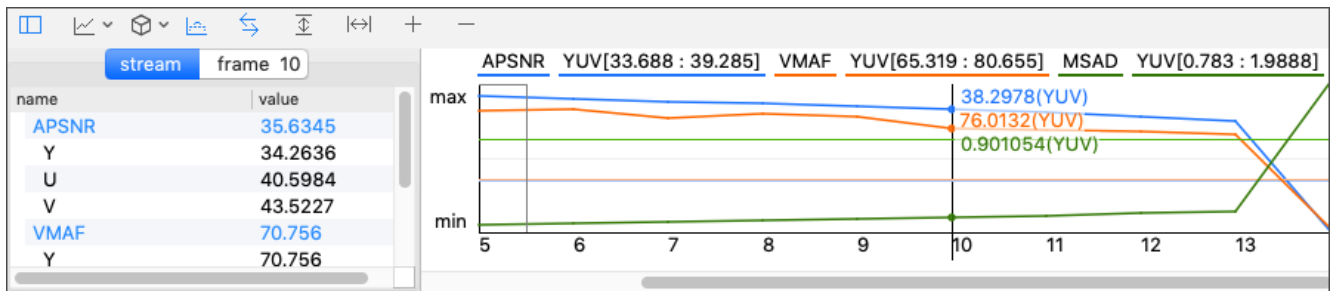


Figure 12. Elecard YUV Viewer – Metrics Window – Local Average Calculation

3.3 Hot Keys

For tabs, functions and modes switching press hot keys described hereto.

Table 7. Hot Keys

Keys combination	Definition
⌘ + O	– opens media files.
⌘ + Shift + O	– opens second media files.
⌘ + R	– opens the recently opened task.
⌘ + ⬆ + R	– opens the second recently opened task.

The following hot keys combinations are used for **Navigation**:

⌘ + ⬆ + Left	– starts the go to beginning function
⌘ + [	– starts the step backward function
⌘ + ⬆ + Space	– starts the play function
⌘ + Space	– starts the step forward function
⌘ +]	– starts the go to end function

To change **View** configuration of panel controls, use the following hot keys:

F3	– shows all controls
F4	– hides all controls

For configuring the **Zoom panel**, use the following hot keys combinations:

⌘ + 1	– shows / activates the stream A panel
⌘ + 2	– shows / activates the stream B panel
⌘ + 3	– shows / activates the frame panel

The following hot keys are used for switching **Output Panels**:

⌘ +1	– activates the A panel
⌘ +2	– activates the B panel
⌘ +3	– activates the DIFF panel

4. Command Line Tool

Elecard YUV Viewer Command Line tool is designed for comparison of two uncompressed video sequences via difference tool (Compare, Subtraction, Temperature, PSNR, PSNR Clip modes) and calculation objective quality metrics, such as PSNR, APSNR, MSE, MSAD, Delta, SSIM, NQI, VIF, VQM, VMAF and VMAF phone.

Elecard YUV Viewer Command Line tool allows automatic analysis of video arrays. You can customize a sequence of commands to perform your specific tasks and automate your routine tasks.

Get to the command prompt by clicking Start->Run and type 'cmd'. Then enter the command to start the application first and proceed with the command line options. In general, the syntax is as follows:

```
./yuvviewer-cli config.xml /inA:inputA.yuv /inB:inputB.yuv /out:output
```

Note: You should not relocate the <product> console without all the other applications of the product contained in the installation pack, otherwise the console application will not start.

4.1 Configuration File

Configuration file – contains all the commands performed by the application and reflects overall operation results:

```
<YUV version = "1">
  <inputA file = "..." type = "IYUV(default)/[*Upsampler Type]" filter =
    "None(default)/FieldGen/Median/Sobel" width = "auto(default)/..." height =
    "auto(default)/..."
    bitdepth = "8(default)/..." offset = "0(default)" step = "1(default)"/>
  <inputB file = "..." type = "IYUV(default)/[*Upsampler Type]" offset =
    "0(default)" step = "1(default)"/>
  <output separator = ".(default)/," delimiter = ";(default)/," file = "..."/>
  <range span = "all(default)/start - stop"/>
  <info>
    <metrics type =
      "PSNR(default)|APSNR|SSIM|DELTA|MSAD|MSE|NQI|VIF|VQM|VMAF|VMAF phone"
      fileType="CSV(default)/JSON" file = "..."/>
    <difference type = "Compare/Subtraction/Temperature/PSNR/PSNR Clip"
      format = "IYUV(default)/YV12" rect_size = "16(default/..)" file = "..."/>
    <data source = "A|B" format = "IYUV(default)/YV12" file = "..."/>
  </info>
</YUV>
```

* **Upsampler types** that can be used in a configuration file: I444 / I422 / IYUV / NV12 / NV21 / P420 / P444 / RGB24 / RGB32 / UYVY (Y422 / UYNV / HDYC) / V210 / V400 / V444 / Y42B / YUY2 (YUNV / V422 / YUYV) / YV12 / YV16 / YVYU / NV21 / V444 .

The command line arguments allow re-defining parameters used in a default configuration file. See examples of the arguments below:

- /inA:<path> - first input file path (inputA file)
- /inB:<path> - second input file path (inputB file)
- /out:<path> - output files prefix (output file)
- /sfx:<string> - suffix of all output files in a configuration file (adds a suffix to an output file before file extension)

- /dc - use current run directory as default path for files
- /? - advanced help

4.2 List of Commands

List of commands contains information on the commands and all parameters, which may be used within a command, and their description.

Table 8. List of Commands

Command	Description
<inputA file> type filter width height bitdepth offset step	The tag is used to configure the input A file; Specifies the type of the YUV data to be analyzed. IYUV is set by default/*Upsampler type Selects the preprocessing filter; Specifies the width of the frame for the file to be analyzed; Specifies the height of the frame for the file to be analyzed; Allows a user to specify values for some metric Types. For several Types, the value is not editable. Available bitdepth types: 8, 9, 10, 11, 12, 13, 14, 15, 16; Starting frame for further calculation. Offset from the beginning up to the specified frame in the stream. It is expressed in frames. Coefficient used for calculation of a successive frame number for further analysis. "1" is set by default.
< inputB file /> type offset step	The tag is used to configure the input B file; Defines the format of the raw data (IYUV (by default) /* Upsampler type; Offset from the beginning up to the specified frame in a reference stream. It is expressed in frames. "0" is set by default. Coefficient used for calculation of a successive frame number for further analysis. "1" is set by default. <i>Note: if the VMAF and VMAF phone metrics are enabled, and if input raw file is missing, encoded reference file should be placed in the input 2 (B) field.</i>
<output/> separator delimiter file	The tag is used to configure the output file; Allows selecting a decimal separator, the "." sign is set by default; Allows selecting data delimiter ";", sign is set by default; Sets the output file path.
<range/> span	The tag is used to define the range of analyzed frames; Defines the range of frames (all (by default) /start - stop), 'start' - number of the first analyzed frame, 'stop' - number of the last analyzed frame.
<metrics/> type file	The tag is used to output information on quality metrics; Specifies the metric type: PSNR (by default) APSNR MSE MSAD DELTA SSIM NQI VIF VQM VMAF VMAF Phone; Sets the output file path where information on metrics will be saved.
<difference/> type format rect_size file	The tag is used to output information on metrics difference Defines the type of difference (compare/subtraction/temperature/psnr/psnr clip). This is the required parameter for this tag. Its default value should be set. If the value is not set, this tag will be skipped; Defines the format for saving data (IYUV (by default)); A block size used to calculate metrics PSNR/PSNR Clip Sets the output file path where information on difference will be saved.
<data/> source format file	The tag is used to save output decoder information on any decoding step; Defines the type of the source for metrics calculation: A B Defines the format for saving data (IYUV (by default)); Sets the output file path where the data will be saved.

4.3 Features of Command Line Tool

- Analysis the whole array of video files automatically;
- Addressing specific and complex challenges with a few lines of commands;
- Retrieving submitted results as a text file in CSV format for easy viewing in Excel;
- Customizing a sequence of commands to perform your specific tasks;
- Accelerating and automating your regular tasks;
- Getting access to most of StreamEye functionality and resources through StreamEye Console.